

Research Methodology: Class5-6



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Class-5: Secondary data collection methods

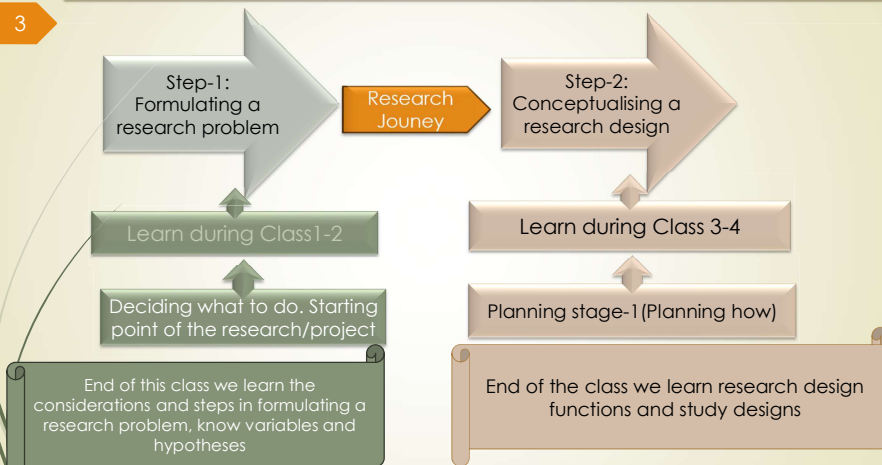
Class-5: Secondary data collection methods

Class-6: Qualitative methods of data collection

Data Collection

Recap from Class 1-2 and class 3-4

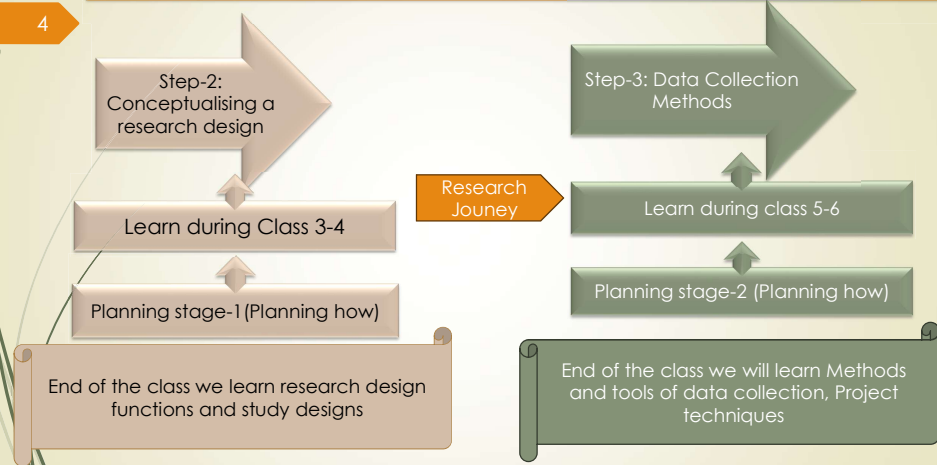
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Assignment-1: Now you must plan for selecting the project area and protect title

Recap from Class 3-4 and Connection to class 5-6

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Assignment-2: Complete planning for your project mentioning what you are going to do

Learning objective: Class-5

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By the end of this class the student should be able to learn and answer the following questions.

- What is primary and secondary sources of data and how these data are different?
- What are the major benefits and limitations of primary and secondary data?
- What are the major criteria or quality checks for the secondary data collection to use to the present study or problem?
- What are the various types and sources of primary and secondary data?

What is data in research?

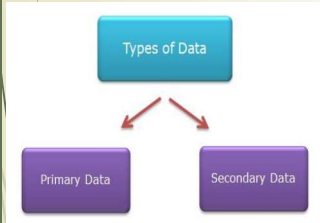
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- Piece of information** about the phenomenon under study.
- In research methodology, data **refers to the facts, figures, observations, or information** collected by a researcher to answer research questions and achieve the objectives of a study.
- Data is the **basic material used for analysis, interpretation, and drawing conclusions** in research.

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What are the two types of research data?



Data in research methodology is mainly classified into:

- Primary Data:** Data collected directly by the researcher for the specific research study.
 - Example: Interviews, questionnaires, surveys, observations, and experiments.
- Secondary Data:** Data that has already been collected by other people or organizations and is used by the researcher.
 - Example: Government reports, census data, books, journals, research papers, and websites.

Data is the **foundation of research**, because researchers use collected information to analyse a problem and reach meaningful conclusions.

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PRIMARY DATA VERSUS SECONDARY DATA

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PRIMARY DATA	SECONDARY DATA	
Primary data is a type of data researchers directly collect from main sources	Secondary data refers to already existing data produced by the previous researchers	Collection method
Includes real-time data	Related to the past	Real-time or past data
Collected to address a current research problem	Primarily collected to address previously existed research problems and can be used to address the current research problem as well	Research problem
Accessing primary data includes a relatively long process	Referring to secondary data is quick and easy	Fast or slow
Data collection tools include observations, surveys, questionnaires, physical testing, online questionnaires, personal or telephone interviews, case studies, and focused group discussions	Data collection tools include journal articles, websites, books, government publications, records, etc.	Tools

Case-study-4.1: Riyas Cafe Research

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Riya wanted to open a healthy-food café near a college. Before investing her money, she decided to conduct research to find out what students actually wanted.

Primary Data: "Data I Collect Myself"

First, Riya went directly to the students.

She prepared a questionnaire and asked 100 college students questions such as:

- What type of food do you prefer?
- How much are you willing to spend on lunch?
- How often do you eat outside?
- Would you prefer healthy food options?

This is PRIMARY DATA.
Because:
"Riya herself went out and collected the information."

She also interviewed 20 students and observed what kinds of food students were buying from existing cafes.

All this information was collected first-hand by Riya for her own research purpose.

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Secondary Data — "Data Already Available"

Riya also wanted to understand the bigger market. Instead of collecting everything herself, she looked at information that **already existed**.

She found:

- Government reports about the local population
- College enrolment statistics
- Published research papers about student food preferences
- Newspaper and magazine articles about the food industry
- Industry reports about café and restaurant trends
- Previous surveys conducted by other researchers

Riya did not collect this information herself. She **used information that had already been collected by others**.

This is **SECONDARY DATA**.
Because:
"Someone else collected it, and Riya not collected herself but using it."

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Essence of the story:

Primary Data = "I collect directly myself." Primary data told her what her potential customers actually thought and wanted.

Secondary Data = "I use what is already collected." Secondary data helped her understand the larger market, existing trends, and background information but data was collected by some other sources.

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Research applications of secondary data

- 1. Problem Identification and Formulation:** Secondary data helps the researcher understand the research problem and develop a conceptual framework for the investigation.
- 2. Hypothesis Designing:** Earlier research, market data, industry trends, and market facts can help the researcher develop assumptions that can be converted into testable hypotheses.
- 3. Sampling Considerations:** Secondary databases containing respondent statistics and contact details can help the researcher define the population and select an appropriate sample.

4. Designing the Primary Research Instrument: Previous studies and available information can be used as a basis for designing questionnaires or other instruments for primary data collection.

5. Validation and Authentication: Earlier records, studies, and data sources can be used to support, verify, or validate findings obtained from primary data.

What are the benefits of primary data

Because the data is collected by the researcher directly with the control in his hand

- **Specific to the research problem:** Primary data is collected according to the exact objectives and requirements of the study.
- **More relevant:** Since the researcher collects the data for a particular purpose, it is usually more relevant to the research question.
- **Greater control over data collection:** The researcher can decide **what information to collect, from whom, when, and how.**
- **Up-to-date information:** Primary data provides current information because it is collected directly at the time of the research.

- **Better accuracy:** The researcher can design appropriate questions and procedures to improve the accuracy and quality of the information.
- **Confidentiality:** The researcher can maintain control over the collected information and decide how it will be used.
- **Can provide detailed information:** Methods such as interviews and observations can provide information that may not be available in existing sources.
- **Useful for decision-making:** Because the data is collected for a specific purpose, it can provide useful evidence for making decisions or solving a particular problem.

Because the data is collected by the researcher directly with the control in his hand

What are the drawbacks of primary data

- **Costly:** Collecting primary data can require money for travel, surveys, interviews, staff, equipment, and other resources.
- **Time-consuming:** Designing the research, selecting respondents, collecting information, and analyzing the data can take a lot of time.
- **Requires skilled researchers:** Proper data collection requires trained people who can prepare questions, conduct interviews, observe accurately, and record information correctly.
- **Difficult to collect from a large population:** Collecting information from a very large number of people can be difficult and expensive.

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Continue....

- **Possibility of bias:** The researcher or respondents may introduce personal opinions or preferences, which can affect the accuracy of the data.
- **Non-response problem:** Some selected respondents may refuse to participate or may not provide complete information.
- **Limited coverage:** Due to limitations of time, money, and resources, the researcher may collect data from only a limited geographical area or group.
- **Data quality may vary:** If respondents misunderstand questions or provide incorrect information, the quality of the collected data may be affected.

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What are the benefits of secondary data

1. Resource Advantage

Secondary data provides a major **saving in both time and cost**. Since the information has already been collected and compiled, the researcher does not have to conduct a completely new data-collection exercise. The time saved can instead be used for **analysis and interpretation**.

2. Accessibility of Data

Secondary data is generally **easier to access** because it has already been collected and organized in a structured form. Government publications, census data, commercial databases, reports, and other sources can be accessed for research purposes.

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What are the benefits of secondary data

3. Accuracy and Stability of Data

Data collected by **recognized organizations or agencies on a large scale** may have the advantage of greater accuracy and reliability. Such data can provide a stable basis for interpretation and decision-making.

4. Assessment of Data

Secondary data can be used to **compare and support the findings of primary research**. It allows researchers to assess whether their primary findings are consistent with information obtained from existing sources.

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What are the drawbacks of secondary data

1. Applicability of Data

The main problem is that secondary data was **originally collected for a different research purpose**. Therefore, it may not be completely suitable or relevant to the objectives of the current study.

There may be differences in:

- Variables being studied
- Units of measurement
- Time period
- Research environment
- Socio-cultural, climatic, or political conditions

These differences can affect the usefulness of the data and may influence the research findings.

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What are the drawbacks of secondary data?

2. Accuracy of Data

The **credibility and reliability of the source** is an important concern when using secondary data. The researcher needs to consider:

- **Who** collected the data?
- **Why** was the data collected?
- **How** was the data collected?
- **What** sampling method was used?
- **How** reliable was the original research process?

Data collected by a reputable organization using an appropriate research methodology is more likely to be reliable.

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Evaluation of secondary data: Research authentication

- There are four evaluation checks are there for the data to be used for the specified problem formulated and research design.
- **These evaluation measures can be employed before one decides to use it for the present study.** The checks are
 - **Methodology Check**
 - **Accuracy Check**
 - **Topical Check**
 - **Cost-benefit Analysis**

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Evaluation of secondary data: Research authentication

1. Methodology Check

This means checking **how the original data was collected**.

The researcher should examine:

- **How** was the study designed?
- **How** were respondents selected?
- **Was** the sample appropriate?
- **Was** there any bias in respondent selection?
- **Were** suitable research methods used?

This means
*"Was the data
 collected using a
 proper research
 method?"*

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Evaluation of secondary data: Research authentication

2. Accuracy Check

This means checking whether the information is correct and reliable.

The researcher should consider:

- **Who** collected the data?
- **Was** the source trustworthy?
- **Was** the information properly analyzed?
- **Is** the data reliable enough for the present study?

This means
"Can I trust this data?"

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Evaluation of secondary data: Research authentication

3. Topical Check

This means checking whether the secondary data is **current and relevant to the present research**.

The researcher should consider:

- How old is the data?
- Does the time period match the present study?
- Have circumstances changed since the data was collected?
- Is the information relevant to the current research problem?

This means
"Is this data recent and relevant?"

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Evaluation of secondary data: Research authentication

4. Cost-Benefit Analysis

The researcher should compare the cost and effort involved in obtaining secondary data with the benefits it provides.

The researcher should ask:

- How much does the data cost?
- How much time will it save?
- Is the data sufficiently useful for the research?
- Is purchasing or using the data worthwhile?

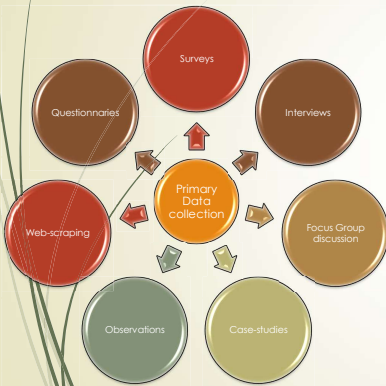
This means
"Is the data worth the cost and effort?"

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Classification of the primary data

Primary data can be broadly classified according to **the way in which it is collected**:

1. **Qualitative Data:** This consists of **non-numerical information** such as opinions, attitudes, feelings, perceptions, experiences, and motivations. It is generally collected through **interviews, focus groups, and observation**.
2. **Quantitative Data:** This consists of **numerical information** that can be measured and statistically analysed. It is generally collected through **surveys, questionnaires, experiments, and measurements**.



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Classification of the secondary data

1. Internal Sources

These are sources within the organization.

Examples:

- Sales records
- Financial statements
- Customer records
- Employee records
- Previous research reports
- Company documents
- Internal reports

This means:
Data already available inside the organization.

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These are sources outside the organization. They are broadly classified as three types:

A. Published Sources: Information that has already been published and made available to researchers.

Government sources:

- Census reports
- Economic Survey
- Statistical publications
- Government department reports
- National Sample Survey data

Non-government sources:

- Books and Journals
- Newspapers and Magazines
- Trade publications
- Research reports
- Directories and indexes
- Reports of international organizations
- Simple meaning: Data that is published for public or research use.

This means:

Data that is published for public or research use.

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Continue...

B. Syndicated Sources

Data collected by specialized research agencies and made available to several organizations or clients.

Examples:

- Retail audits
- Wholesale audits
- Consumer/household panels
- Purchase panels
- Media panels
- Television Rating Performance (TRP) data
- Simple meaning: Data collected by professional research agencies and provided to multiple users.

This means:

Data collected by professional research agencies and provided to multiple users.

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Continue....

C. Electronic Sources

Secondary information available through electronic and online media.

Examples:

- Online databases
- Research databases
- Electronic journals
- Websites
- Online reports
- Digital publications

This means:

Data that can be accessed electronically or through the internet.

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Class-6: Qualitative methods of data collection

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Learning Objective: Class-6

By the end of this class the student should be able to learn and answer the following questions.

- **What do you mean by quantitative and qualitative research?**
- **Elaborate the basic premise for using qualitative research method?**
- **Differentiate between quantitative and qualitative data methods?**
- **What is focus group discussion in research?**

What is Quantitative Research

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Here the data can be quantified by numerical values such as age, money, etc.

- **Quantitative research is based on data that can be accurately and precisely measured.**
- This approach is also known as **empirical research**.
- Quantitative or empirical research is **statistical in nature** and quantifies opinions, behaviours, attitudes and other defined variables.
- It provides **measurable data** to formulate theories and facts and uncover patterns.
- The data collected from quantitative research can be transformed into usable statistics.
- **Example:** Survey 500 students to measure their satisfaction with online learning (**quantitative**).

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What is qualitative research?

Here the data cannot be quantified and not measurable such as feelings, satisfaction level, likeness level, etc.

- **Qualitative Research is exploratory and investigative in nature.**
- **It involves data that can be observed, but not measured. Thus, it classifies the qualities of what is studied.**
- Qualitative research is used to understand the underlying reasons, opinions, and motivations behind something as well as to uncover trends in thought and opinions.
- This approach can be used to analyze a problem in depth and develop hypothesis or theories.
- **Example:** Interview 20 students to understand the reasons behind their satisfaction or dissatisfaction (**qualitative**).

Premise for using qualitative research methods

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Means which type of problem or conditions we will prefer to do investigation through the qualitative research methods

Qualitative research is used when the researcher wants to understand a problem deeply rather than simply measure it with numbers.

The main premises are:

- **Human behavior is complex:** People's actions, thoughts, feelings, and experiences cannot always be explained through numbers alone.
- **Meaning is important:** Researchers may need to understand what people think, feel, believe, and why they behave in a particular way.
- **Context matters:** People's behavior can be influenced by their social, cultural, organizational, and environmental surroundings. Qualitative methods help study behavior within its real-life context.

Premise for using qualitative research methods

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Means which type of problem or conditions we will prefer to do investigation through the qualitative research methods

- **Research questions may be exploratory:** When little is known about a topic, qualitative research helps the researcher explore and discover new ideas, concepts, and explanations.
- **Participants' perspectives are valuable:** Qualitative research gives participants an opportunity to explain their experiences in their own words.
- **Reality can have different interpretations:** Different people may experience the same situation differently. Qualitative research tries to understand these different viewpoints and interpretations.

Difference Between Quantitative and Qualitative Research

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The difference between qualitative and quantitative data methods can be explained in simple sentences:

1. **Qualitative research deals with non-numerical data**, such as words, opinions, feelings, experiences, and observations, whereas **quantitative research deals mainly with numerical data** that can be measured and statistically analysed.
2. **Qualitative methods are used to understand the meaning, reasons, and experiences behind a situation**, whereas **quantitative methods are used to measure the extent, frequency, or magnitude of a situation**.
3. **Qualitative research generally answers questions such as "why," "how," and "what,"** whereas **quantitative research generally answers questions such as "how much," "how many," and "to what extent."**

Easy way to remember

Qualitative → Words → Meaning → Why/How → Deep understanding

Quantitative → Numbers → Measurement → How many/How much → Statistical analysis

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4: **Qualitative methods are usually flexible and exploratory**, allowing the researcher to investigate a topic in depth, whereas **quantitative methods are generally structured and predetermined**.

5: **Qualitative data is commonly collected through interviews, focus groups, observations, and case studies**, whereas **quantitative data is commonly collected through structured questionnaires, surveys, experiments, and measurements**.

6: **Qualitative research usually involves a relatively small sample** and provides detailed information about participants, whereas **quantitative research generally uses a larger sample to obtain measurable and comparable results**.

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7: **Qualitative data is analyzed by identifying themes, patterns, concepts, and meanings**, whereas **quantitative data is analyzed using statistical techniques such as averages, percentages, correlations, and tests of significance**.

8: **Qualitative research focuses more on depth and understanding**, whereas quantitative research focuses more on measurement, precision, and generalization.

9: **In qualitative research, the researcher often plays an active role in interpreting participants' views and experiences**, whereas in **quantitative research, the researcher generally follows standardized procedures to maintain objectivity**.

QUANTITATIVE RESEARCH VERSUS QUALITATIVE RESEARCH

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Quantitative Research uses statistics	Qualitative Research uses descriptions and observations	Statistics
Data can be measured accurately	Data can be observed and not measured	Accuracy
Considered to be objective	Considered to be subjective	Objective
Uncovers measurable data to formulate theories and facts and uncover patterns	Helps to understand the underlying reasons, opinions, and motivations	Data
Mainly uses hypotheses	Uses either hypotheses or research questions	Hypothesis
Data collection methods are highly structured	Data collection methods are semi-structured or unstructured	Data collection method
The sample population is large	The sample population is small	Sample

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Methods of Quantitative Research

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- **Survey Method:** The researcher collects information from a large number of people using a structured questionnaire or schedule. The responses are converted into numerical data and analyzed statistically.
- **Experimental Method:** The researcher studies the effect of one variable on another under controlled conditions. For example, a researcher may introduce a new teaching method and measure whether students' performance improves.
- **Observation Method:** The researcher systematically observes and records specific behaviors or events. The observations are usually recorded in a structured and measurable form so that they can be analyzed numerically.
- **Ex-post Facto Method:** The researcher studies an existing situation or event after it has occurred. The researcher does not manipulate the variables but examines possible relationships between them.

Methods of Quantitative Research

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Survey → collects numerical information from many people.

Experiment → tests cause and effect under controlled conditions.

Observation → measures and records behaviour.

Ex-post facto → studies events that have already happened.

Correlation → studies relationships between variables.

Causal-comparative → compares existing groups to identify possible causes of differences.

■ **Correlational Method:** This method is used to determine whether two or more variables are related to each other. For example, a researcher may examine the relationship between study time and examination marks.

■ **Causal-Comparative Method:** The researcher compares two or more existing groups to find possible reasons for differences between them. The researcher does not directly manipulate the independent variable.

Methods of qualitative research

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- **In-depth Interview:** The researcher has a detailed conversation with an individual to understand their thoughts, feelings, experiences, and opinions about a particular issue.
- **Focus Group Discussion:** A small group of people is brought together to discuss a topic. The researcher observes the discussion and tries to understand the different views and opinions of the participants.
- **Observation:** The researcher observes people's behaviour and activities in their natural setting. This helps the researcher understand what people actually do rather than relying only on what they say.

Methods of qualitative research

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Interview → asks people directly

Focus group → discusses with a group

Observation → watches behaviour

Case study → studies one case deeply

Projective techniques → explores hidden feelings and motivations

Content analysis → studies documents or communication

- **Case Study:** The researcher studies a particular person, group, organization, event, or situation in detail. Information is collected from different sources to develop a complete understanding of the case.
- **Projective Techniques:** Used to understand people's underlying feelings, attitudes, or motivations. Participants are given an indirect task or situation and are asked to respond to it, allowing the researcher to explore thoughts that may not be expressed directly.
- **Content Analysis:** The researcher systematically examines written, spoken, or visual material, such as documents, interviews, advertisements, or media, to identify important themes, ideas, and patterns.

Suppose your research topic is: "A qualitative study of students' experiences and perceptions of college life." Instead of directly asking, "Are you happy with your college life?", the researcher uses a **projective technique**. The question may be "Tell your college life experience".

Focus group discussion in research

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FOCUS GROUP RESEARCH

PURPOSE
gather insights through group discussions

FOCUS GROUP PROCESS
1. DEFINE OBJECTIVES 2. RECRUIT PARTICIPANTS 3. CONDUCT DISCUSSION 4. ANALYZE RESULTS

ROLES
MODERATOR PARTICIPANTS

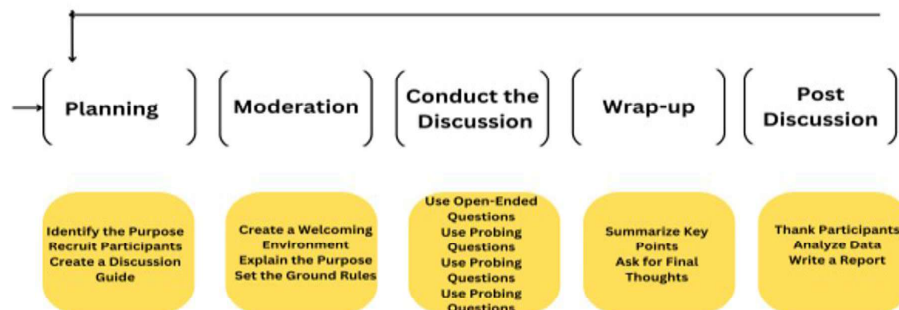
BENEFITS
qualitative feedback
explore attitudes & perceptions
generate new ideas
inform decision-making

- A **Focus Group Discussion (FGD)** is a qualitative research method in which a small group of people discuss a specific topic under the guidance of a trained moderator.
- It is used to gather in-depth information about participants' opinions, attitudes, beliefs, experiences, and perceptions.
- The key elements of a focus group are: Size of the group, Nature of the members, Time period, The recoding and The moderator, etc.



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Planning/Process for Focus Group Discussion



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Factors for selection of appropriate method for data collection for our problem



Thanks